

HW1 Individual Term Project Proposal

Emerging Technologies and Applied Innovation, Fall 2026

1. Project Title

Self-Calibrating UWB-to-Video Fusion: Persistent Player Identity and Jump Clips from Any Camera

2. Primary Project Platform

LocusConnect - Spatial Intelligence

3. Problem and Target User

Coaches and analysts at university and club teams review video to judge what each player did, but video trackers lose identity at crossings: the best published trackers still leave about 2,600 identity switches on SportsMOT. Analysts fix these by hand, hours per match, so training sessions go unanalysed. The LocusConnect UWB stream keeps every player's identity and 3D position through those occlusions, yet commercial tools only align such telemetry to video in time, by a manual sync point. Projecting identity onto players in the frame needs a surveyed, fixed camera, so it never happens at sessions filmed from a phone.

4. Proposed Project Idea

Input: video from any camera and the live Locus RTLS 3D stream (tag ID, x, y, z, timestamp). Processing: (1) markerless self-calibration during the first minute of free play, jointly estimating camera pose, focal length, and the camera-to-UWB clock offset by matching detected hip keypoints to 3D tag trajectories, with RANSAC over unknown tag-to-person correspondences; (2) per frame, each tag's 3D position is projected into the image and matched to a detection by gated assignment, so identity survives crossings; (3) as a stretch extension, the Z-axis stream flags jumps above a noise-derived threshold. Output: video with persistent name labels on every tagged player; if the stretch goal is reached, an auto-cut clip of each jump with height overlaid.

5. Technology and Data Required

OpenCV for PnP and joint refinement of pose, focal length, and time offset; YOLO-pose and ByteTrack for keypoints and tracks; Python with Hungarian assignment and a Kalman smoother. The work is 3D geometry, estimation, and integration. Data: synchronized phone video and Locus RTLS 3D logs of three to six tagged people in a gym, with staged crossings and jumps, plus manual identity and jump labels. Not yet existing; recorded in two sessions on the course's LocusConnect installation through its MQTT or WebSocket API, confirmed in week one. A ChArUco marker calibration serves as the reference for the self-calibration metrics.

6. Final Prototype / Live Demo

At the end of the semester, I will be able to demonstrate that a phone camera on a tripod at an arbitrary spot calibrates itself within one minute of four tagged people moving freely, with reprojection error converging on screen. After that, each person carries a persistent name label through repeated crossings; toggling fusion off makes labels swap, toggling it on restores them within a frame. If the stretch goal is reached, a clip appears within seconds of a jump showing it with its height overlaid. If live UWB access fails, the same pipeline runs on a recorded synchronized session.

7. How Will You Know It Works?

Identity: ID switches per minute, ByteTrack alone versus fused, on a labelled session; target 90 percent fewer, and zero on staged crossings; per-frame label accuracy above 95 percent. Calibration: self-calibrated pose versus the marker reference, target rotation error below 1 degree, translation below 20 cm, reprojection below 15 pixels, convergence within 60 seconds, time offset within 20 ms. Jumps (stretch): measured UWB vertical noise (published LPS validations report 17 to 21 cm) sets the threshold; precision and recall above 0.9 for jumps above it, against video labels. Latency below 100 ms per frame.

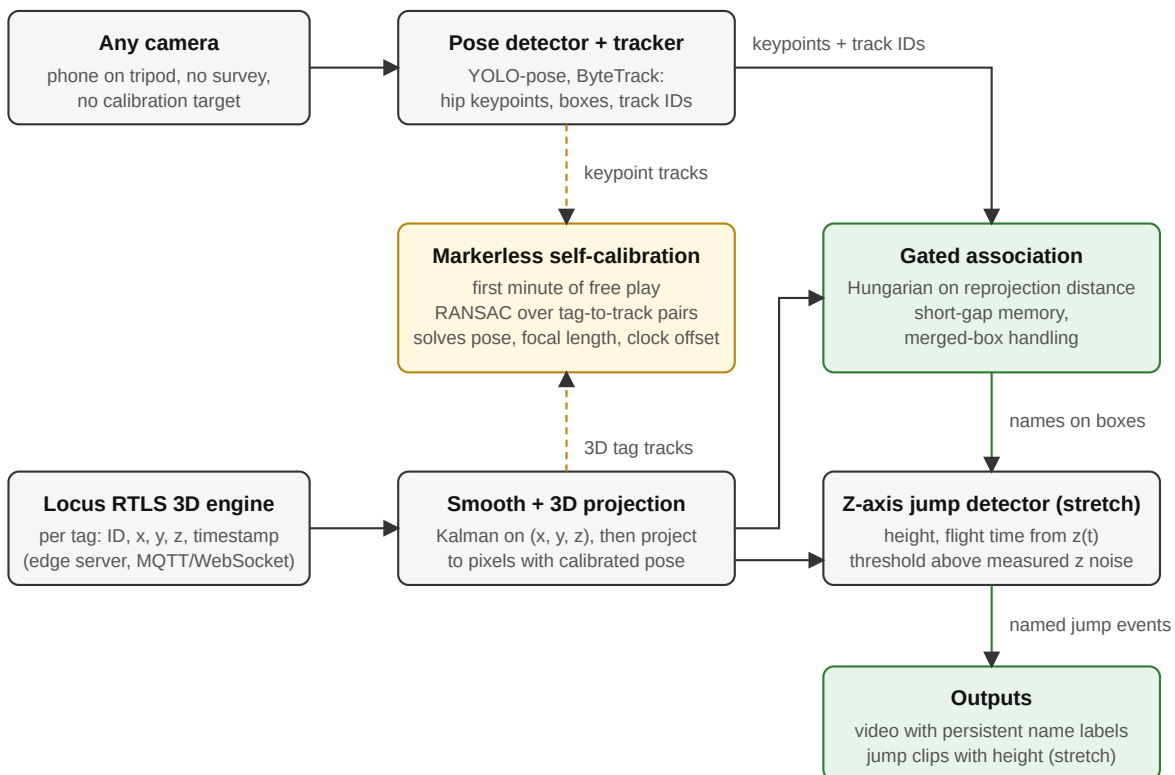
8. Why Should This Project Be Selected?

Identity is the measured bottleneck of sports video tracking, and no published method calibrates an arbitrary camera to UWB from several tagged players with unknown correspondences; that piece combines established methods, so it fits a semester. Practical value for LocusConnect is direct: every coach already films training on a phone, and this puts tag identity into that footage. Stable identity then unlocks per-player clips and statistics.

9. Your Expected Contribution

I work in Python with OpenCV and ROS on multi-sensor robotics data, including camera calibration and sensor time synchronization. I will own the calibration and association pipeline, the data-collection sessions, and the evaluation. I want to develop real-time multi-object tracking and pose-based video analytics.

System Diagram (optional page)



Dashed: one-time setup (calibration outputs the camera pose and clock offset used by the projection block). Solid: per frame.